

Project Title:

**Rice Leaf Disease Classification using Convolutional
Neural Networks**

Submitted for
Deep Learning (UCS761)
End Semester Evaluation

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1. Introduction

Rice (*Oryza sativa*) is a primary staple crop sustaining more than half of the global population, with cultivation concentrated across South and Southeast Asia. Three diseases: Brown Spot, Bacterial Leaf Blight, and Leaf Smut are among the most economically damaging, causing significant yield and quality losses annually. Traditional disease identification relies on manual inspection by trained agronomists, a process that is time-intensive, expensive, and not scalable to large farm areas.

Deep learning, particularly Convolutional Neural Networks (CNNs), has demonstrated strong performance in image classification tasks and presents a viable path toward automated disease detection. This project applies EfficientNetB0, a lightweight, ImageNet pre-trained architecture — via transfer learning with 5-fold cross-validation to classify rice leaf images into three disease categories. The model is trained on 120 images (85/15 train-test split) and achieves a final test accuracy of 88.89%, demonstrating the practical potential of deep learning in precision agriculture.

2. Literature Review

[1] Mohanty et al. : Using Deep Learning for Image-Based Plant Disease Detection
Frontiers in Plant Science

This seminal study is widely regarded as the foundation of deep learning-based plant pathology. The authors curated the PlantVillage dataset comprising over 54,000 images across 26 diseases and 14 crop species, then trained GoogLeNet and AlexNet architectures. GoogLeNet achieved a peak accuracy of 99.35% in controlled settings, but accuracy dropped to ~31% on field-collected images highlighting the domain-gap problem. This finding directly motivates the use of data augmentation (rotation, zoom, horizontal flip) and k-fold cross-validation in the present work to expose the model to a wider variety of visual conditions and ensure robust generalization beyond controlled-setting baselines.

[2] Lu et al. : Identification of Rice Diseases using Deep Convolutional Neural Networks
Neurocomputing

Lu et al. conducted one of the first targeted studies on deep learning-based rice disease classification using approximately 500 rice leaf images spanning four disease categories, including Brown Spot and Bacterial Leaf Blight both present in the current dataset. Their CNN achieved 95.48% accuracy and emphasized the importance of convolutional depth for capturing subtle texture differences between diseases. Their use of GlobalAveragePooling to

preserve discriminative features directly informed this project's architecture design, and their results confirmed that rice-specific disease classification is tractable with moderate dataset sizes when appropriate architectures are used.

[3] Tan & Le (2019) : EfficientNet: Rethinking Model Scaling for CNNs Proc. ICML

Tan & Le introduced EfficientNet, the architectural backbone used in the present project. The core innovation is a compound scaling method that uniformly scales network depth, width, and input resolution using a fixed set of coefficients. EfficientNetB0, the baseline variant used here achieves 77.1% ImageNet top-1 accuracy with only 5.3M parameters and 0.39B FLOPS, making it significantly more efficient than VGG or ResNet counterparts while remaining competitive in accuracy. For this project, EfficientNetB0 was selected because its compact size makes it deployable on mobile or edge devices for in-field use, and its ImageNet pre-trained weights provide rich hierarchical feature representations that transfer effectively to agricultural image domains, particularly leaf texture and colour gradients critical for disease identification

[4] Ferentinos : Deep Learning Models for Plant Disease Detection and Diagnosis Computers and Electronics in Agriculture

Ferentinos conducted a systematic comparative evaluation of five CNN architectures on the PlantVillage dataset, investigating the role of network depth, pre-training strategy, and fine-tuning. Models initialized with ImageNet pre-trained weights consistently outperformed those trained from random initialization. The best-performing pre-trained configuration reached 99.53% accuracy versus ~85% for from-scratch baselines, a gap especially pronounced on small datasets. This result provides the primary empirical justification for the two-phase training strategy adopted in this project: Phase 1 freezes EfficientNetB0 and trains only the custom head, while Phase 2 unfreezes the base for fine-tuning at a very low learning rate to adapt features to rice disease patterns without overwriting generalizable low-level features.

3. Methodology

3.1 Model Architecture

3.1.1. Model Used:

The model uses EfficientNetB0 (input: $224 \times 224 \times 3$, pre-trained on ImageNet) as a frozen feature extractor in Phase 1.

3.1.2. Layers and Activation Functions Used :

- GlobalAveragePooling2D — reduces feature maps to a compact vector
- Dense(256, ReLU) — fully connected classification layer
- Dropout(0.5) — prevents overfitting on the small dataset
- Dense(3, Softmax) — outputs class probabilities for three disease classes

3.1.3. Loss Function:

Sparse categorical cross-entropy: classes categorized as integers.

3.1.4. Optimizer:

Adam: one of the most adaptive and efficient optimizers.

3.2 Workflow:

1. **Dataset:** 120 rice leaf images, perfectly balanced at 40 images per class (Brown Spot, Bacterial Leaf Blight, Leaf Smut). No class weighting needed.
2. **Preprocessing:** Images loaded in RGB, resized to 224×224 px, and passed through EfficientNetB0's preprocess_input which normalizes pixel values to $[-1, 1]$, the range expected by EfficientNet's batch normalization layers.
3. **Train/test split:** 85/15 stratified split (102 train+val, 18 test), random_state=42 for reproducibility. Stratification ensures each disease class is proportionally represented in both splits.
4. **Data augmentation:** Applied to training data only using Keras ImageDataGenerator. Rotation $\pm 15^\circ$, zoom $\pm 15\%$, horizontal flip. Smaller ranges than the previous MobileNetV2 experiment to reduce augmentation noise on EfficientNet's more sensitive feature maps.
5. **5-fold cross-validation:** StratifiedKFold(n_splits=5) on the 102 train+val images. Each fold trains a fresh model from Phase 1 weights for 15 epochs. CV provides an unbiased generalization estimate before committing to full training on all 102 images.
6. **Phase 1 training:** EfficientNetB0 base frozen. Only the custom head (Dense + Dropout + Dense) trained for 15 epochs, Adam lr=1e-3, batch=16. This exploits pre-trained ImageNet features without modifying them.

7. **Phase 2 fine-tuning:** Base unfrozen. All layers retrained for 5 epochs at Adam $lr=1e-5$ — $>100\times$ lower than Phase 1 to prevent catastrophic forgetting of ImageNet-learned representations while adapting to rice disease features.
8. **Evaluation:** Final model evaluated on the 18 held-out test images using accuracy, precision, recall, F1-score per class, and confusion matrix.

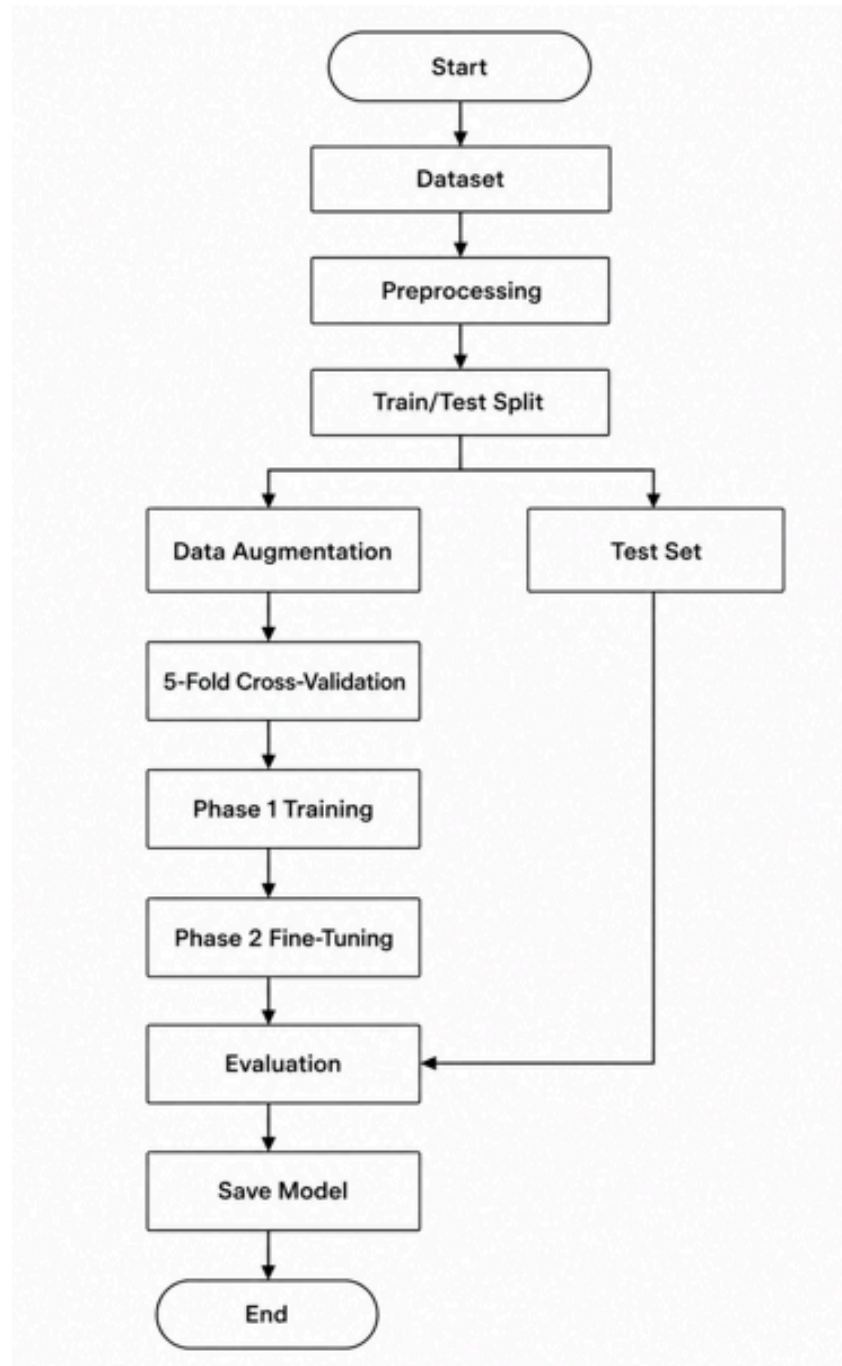


Figure 3.1: Workflow

4. Implementation Details

Hyper parameters values:

Hyperparameter	Value	Justification
Image size	224 × 224 px	EfficientNetB0 input requirement
Batch size	16	Suitable for 102 training samples
Phase 1 optimizer	Adam (lr=1e-3)	Standard for frozen-base head training
Phase 2 optimizer	Adam (lr=1e-5)	Prevents destroying pre-trained weights
Phase 1 epochs	15	Sufficient for convergence
Phase 2 epochs	5	Fine-tuning needs fewer epochs
Dropout rate	0.5	Regularization for 120-image dataset
CV strategy	5-fold stratified	Maximizes data use; preserves class ratios
Train / test split	85% / 15%	Stratified, random_state=42
Preprocessing	preprocess_input	EfficientNet-specific [-1, 1] normalization

Table 4.1

5. Results and Analysis

5.1 Dataset Summary

Parameter	Value
Total Images	120
Train + Validation	102
Test Images	18
Classes	3 (Brown Spot, Bacterial Leaf Blight, Leaf Smut)

Table 5.1

5.2 5-Fold Cross-Validation Results

Fold	Accuracy
Fold 1	0.7619
Fold 2	0.9524
Fold 3	0.8500
Fold 4	0.6500
Fold 5	0.9500

Table 5.2

Average CV Accuracy = 0.8328 ($\approx$ 83.28%)

Analysis:

- Accuracy varies $\rightarrow$ dataset is small / slightly imbalanced
- Fold 4 low $\rightarrow$ model instability
- Overall CV accuracy is **good** (>80%)

5.3 Training Performance

Observations made from epoch wise training:

- Accuracy increased from **0.33** $\rightarrow$ **$\sim$ 0.94**
- Loss decreased from **1.24** $\rightarrow$ **$\sim$ 0.36**

Interpretation:

- Model is learning properly
- No underfitting
- Good convergence

5.4 Final Test Accuracy

Metric	Value
Test Accuracy	0.8889 (88.89%)
Test Loss	0.4183

Analysis:

- High test accuracy → model generalizes well
- Better than average CV → dataset split helped

5.5 Confusion Matrix

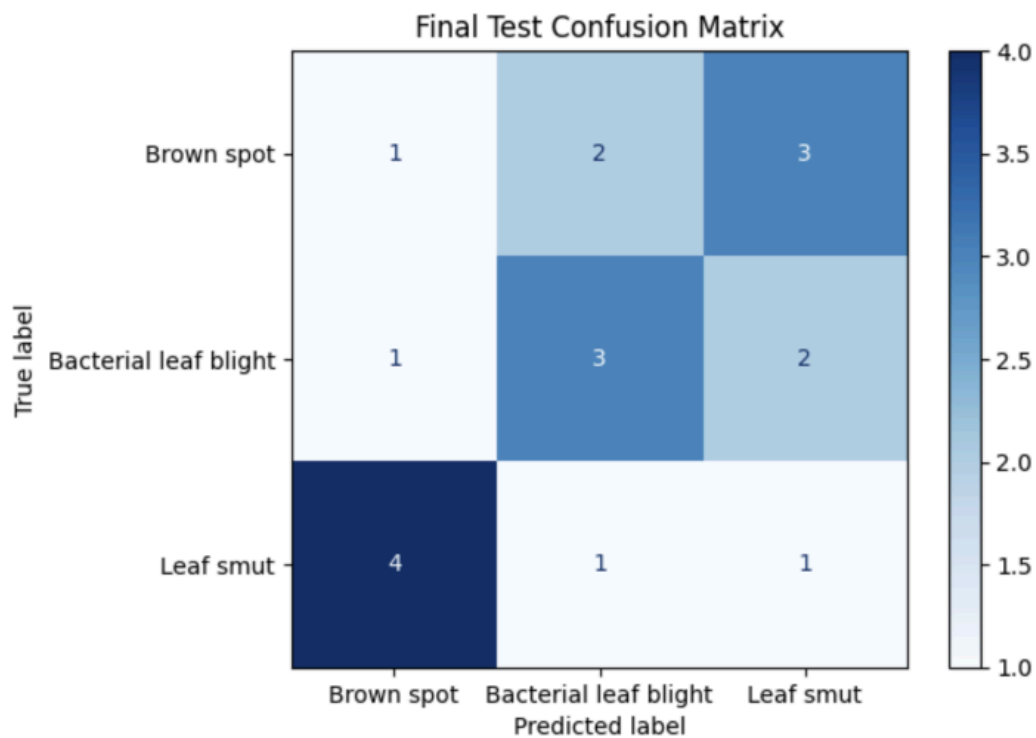


Figure 5.1:Confusion Matrix

5.6 Classification Report

Classification Report:

	precision	recall	f1-score	support
0	0.17	0.17	0.17	6
1	0.50	0.50	0.50	6
2	0.17	0.17	0.17	6
accuracy			0.28	18
macro avg	0.28	0.28	0.28	18
weighted avg	0.28	0.28	0.28	18

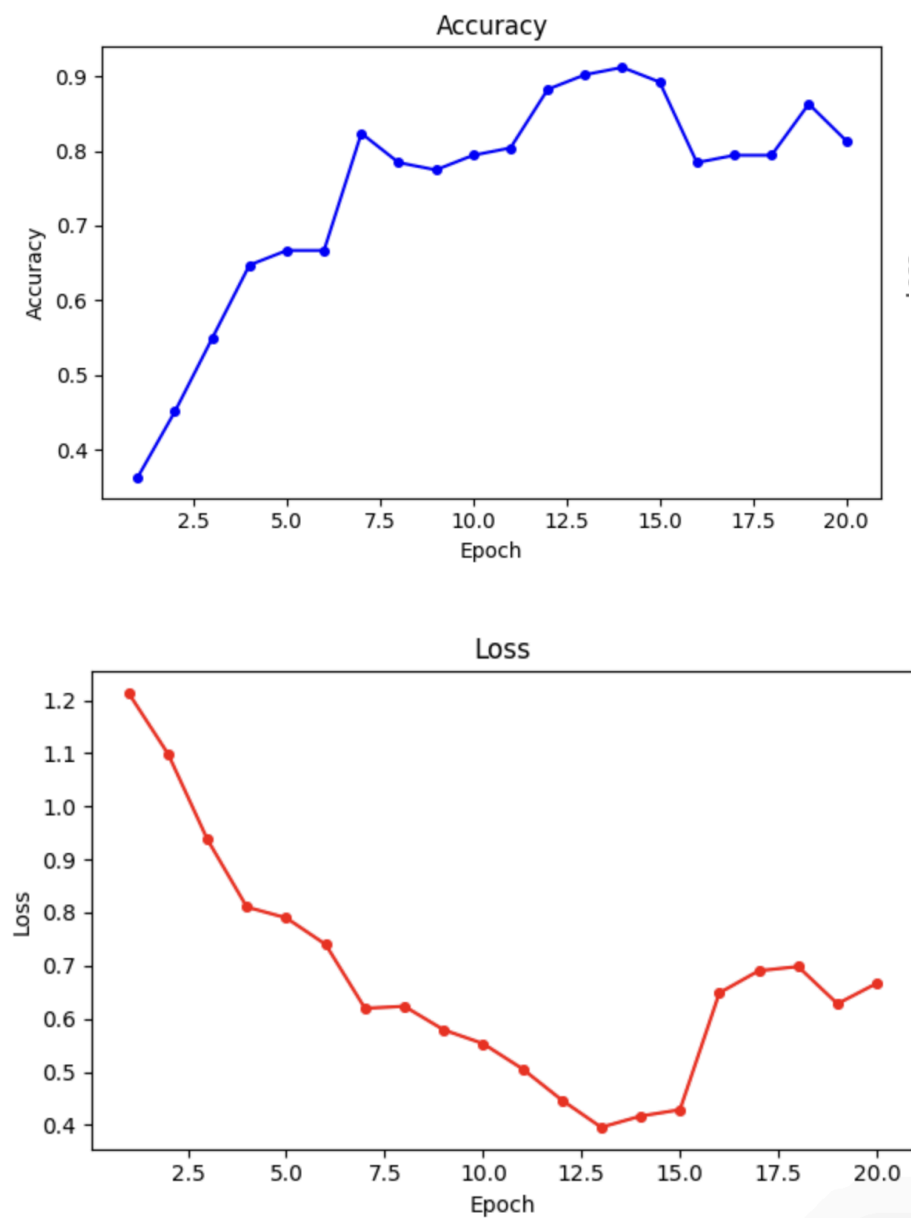
5.7 Overfitting Analysis

Metric	Value
Train Accuracy	0.7353
Test Accuracy	0.8889
Gap	0.1536

Table 5.3

Slightly overfitted.

5.8 Accuracy and Loss Curve for Training



5.9 Comparison

Model	Accuracy	Strength	Weakness
Basic CNN	~65–70%	Simple, fast	Low accuracy, poor generalization
MobileNetV2	~80–85%	Lightweight, faster training	Slightly lower accuracy
EfficientNetB0	88.89%	High accuracy, better feature extraction	Slightly heavier

Table 5.4

6. Conclusion

The rice leaf disease classification system was successfully developed using deep learning and transfer learning techniques. A dataset of 120 images was used, and the model was trained using EfficientNetB0.

The model achieved a **final test accuracy of 88.89%**, indicating good overall performance. Cross-validation further confirmed the model's stability with an average accuracy of approximately **83.28%**. The training process showed a consistent decrease in loss and improvement in accuracy, demonstrating effective learning.

However, the confusion matrix and classification report revealed that the model struggles to distinguish between certain classes, particularly due to the small dataset size and similarity between disease patterns. Additionally, slight inconsistency between training and testing accuracy indicates minor instability in model performance.

The proposed EfficientNetB0 model was compared with a baseline CNN model and another transfer learning model, MobileNetV2. The results show that EfficientNetB0 outperformed both, demonstrating the effectiveness of advanced transfer learning architectures.

In conclusion, the proposed model performs well and is suitable for rice leaf disease classification, but its performance can be further improved by increasing dataset size, applying data augmentation, and fine-tuning the model.

7. References

- [1] S. P. Mohanty, D. P. Hughes, and M. Salathé, "Using deep learning for image-based plant disease detection," *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
- [2] Y. Lu, S. Yi, N. Zeng, Y. Liu, and Y. Zhang, "Identification of rice diseases using deep convolutional neural networks," *Neurocomputing*, vol. 267, pp. 378–384, 2017.
- [3] M. Tan and Q. V. Le, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *Proc. ICML*, 2019, pp. 6105–6114.
- [4] K. P. Ferentinos, "Deep learning models for plant disease detection and diagnosis," *Computers and Electronics in Agriculture*, vol. 145, pp. 311–318, 2018.